

RUIAN, China

WMO: 587520

Lat: 27.7832N

Lon: 120.6489E

Elev: 38

StdP: 100.87

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.0	-9.8	1.6	5.8	-6.7	2.1	6.4	6.0	7.2	5.4	7.7	2.4	290	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	34.2	27.1	33.1	26.9	32.2	26.6	28.1	32.1	27.7	31.4	27.3	30.8	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	30.2	26.7	22.4	29.9	26.3	21.9	29.6	91.0	32.4	88.8	31.6	86.9	30.9	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.4	4.7	-0.2	36.7	1.7	1.0	-1.5	37.4	-2.5	38.0	-3.5	38.6	-4.7	39.3		
			-2.8	29.0	1.8	0.7	-4.1	29.5	-5.1	29.9	-6.1	30.3	-7.4	30.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	9.0	9.7	12.7	17.5	21.8	25.3	29.1	28.9	26.3	21.9	16.9	11.6
	DBStd	7.60	3.38	3.51	3.28	3.07	2.52	2.72	1.74	1.40	2.23	2.51	3.38	3.45
	HDD10.0	141	60	45	11	1	0	0	0	0	0	0	1	23
	HDD18.3	1041	291	242	175	51	3	0	0	0	3	65	211	
	CDD10.0	3526	28	37	96	224	367	460	592	587	488	368	209	72
	CDD18.3	1384	0	0	2	25	111	210	333	328	238	113	23	1
	CDH23.3	12963	0	0	3	59	431	1689	4141	3999	2158	448	34	1
	CDH26.7	4725	0	0	0	7	72	509	1811	1663	608	53	1	0
Wind	WSAvg	2.3	2.1	2.2	2.2	2.0	2.0	2.0	2.5	2.7	2.5	2.5	2.2	2.2
Precipitation	PrecAvg	1691	69	86	141	142	185	251	161	239	198	80	79	60
	PrecMax	2257	175	162	297	240	380	523	382	412	501	334	163	200
	PrecMin	1218	7	16	46	41	75	59	32	19	40	4	6	4
	PrecStd	271	39	39	64	57	80	103	89	113	115	81	46	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	21.1	23.5	27.5	30.8	34.4	35.6	35.5	33.1	29.9	26.3	21.9
		MCWB	15.0	16.6	17.8	22.1	25.4	27.9	27.6	27.2	26.4	24.2	21.1	17.1
	2%	DB	17.3	18.4	21.1	24.8	28.6	32.3	34.5	33.8	31.6	28.1	24.2	19.4
		MCWB	14.0	14.3	16.3	19.8	23.1	27.1	27.4	26.7	25.3	22.3	19.5	15.3
	5%	DB	15.3	16.1	19.3	23.3	27.0	30.8	33.5	32.8	30.7	26.8	22.7	17.7
		MCWB	12.2	12.8	15.2	19.0	22.4	26.4	27.1	26.7	24.9	21.4	19.1	14.0
	10%	DB	13.6	14.3	17.4	22.0	25.6	29.5	32.4	32.0	29.7	25.7	21.6	16.2
		MCWB	10.8	11.8	14.2	18.4	21.6	25.7	26.9	26.4	24.6	20.7	18.4	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	16.9	19.0	23.1	26.2	28.5	28.7	27.6	26.0	22.4	19.1	
		MCDB	18.4	20.4	22.1	25.4	30.0	32.7	33.3	32.4	30.7	27.7	23.9	20.2
	2%	WB	14.6	15.0	17.3	21.6	24.9	27.6	28.2	28.1	26.9	24.9	21.3	16.6
		MCDB	16.9	17.6	19.9	23.4	27.3	31.2	32.5	31.5	29.9	26.8	23.2	18.5
	5%	WB	12.8	13.5	16.0	20.4	23.9	26.8	27.7	27.6	26.3	23.4	20.3	14.8
		MCDB	14.7	15.6	18.1	22.3	26.0	30.0	31.8	30.9	29.1	25.8	22.1	16.8
	10%	WB	11.4	12.2	14.9	19.1	22.8	26.1	27.3	27.2	25.7	21.9	19.1	13.5
		MCDB	13.1	13.9	17.2	21.3	24.6	29.0	31.2	30.5	28.5	24.6	21.0	15.5
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.4	6.1	6.2	5.7	5.1	5.6	5.7	5.5	6.1	5.7	6.2
		MCDBR	8.3	8.8	8.9	8.2	7.7	6.8	7.3	7.0	6.5	7.0	7.0	8.2
		MCWBR	5.1	5.4	5.1	4.4	3.6	3.2	2.8	2.5	2.5	3.1	3.6	4.6
	5% WB	MCDBR	7.2	7.8	7.3	7.2	6.4	6.7	6.5	6.2	6.0	5.4	5.6	6.2
		MCWBR	4.9	5.4	4.8	4.3	3.5	3.3	2.9	2.6	2.6	3.2	3.3	4.3
Clear-Sky Solar Irradiance	taub	0.578	0.634	0.686	0.692	0.642	0.618	0.567	0.594	0.587	0.581	0.565	0.546	
	taud	1.729	1.653	1.589	1.632	1.775	1.859	2.001	1.917	1.900	1.829	1.824	1.810	
	Ebn at Noon	641	646	647	664	700	712	749	725	714	681	650	646	
	Edh at Noon	201	234	264	260	225	206	178	193	191	193	181	178	
All-Sky Solar Radiation	RadAvg	2.26	2.57	3.36	3.86	4.12	4.23	5.83	5.12	4.21	3.65	2.53	2.38	
	RadStd	0.46	0.60	0.33	0.45	0.61	0.52	0.54	0.47	0.42	0.62	0.38	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page